

Dancing on the Saddles: A Geometric Framework for Stochastic Equilibrium Dynamics[†]

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Abstract

Extending saddle-path analysis to stochastic environments, I introduce *conditional saddle paths*: equilibrium trajectories under frozen exogenous regimes, traced by stochastic-equilibrium decision rules. Their *geometry* diagnoses state dependence: shock responses depend on position along the saddle and on slope gaps across regimes. Their *monotonicity* delivers dimension reduction: aggregate capital exactly indexes the equilibrium state within a frozen-regime spell in [Krusell and Smith \(1998\)](#), and a synchronization theorem gives sufficient conditions under which the distributional gap across histories shrinks with their shared recent regime path. Finally, the framework turns policy comparative statics in stochastic general equilibrium into geometry: in a search-and-matching economy, unemployment volatility is hump-shaped in the benefit level, because high benefits pull the regime-specific conditional steady states together, leaving little room to fluctuate.

Keywords: Conditional saddle path, business cycles, state-dependent dynamics, sufficient statistics, heterogeneous agents, search and matching.

JEL codes: C62, D58, E32, J64.

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1 Introduction

Understanding equilibrium dynamics in models with aggregate uncertainty remains a central challenge in macroeconomics. Unlike deterministic models, where saddle-path diagrams provide immediate geometric intuition, stochastic equilibrium models lack comparable geometric frameworks. This makes it difficult to develop intuition about how economies respond to shocks, how different states interact, and why certain computational methods work. The challenge becomes particularly acute in heterogeneous-agent models, where the natural state variable is an infinite-dimensional wealth distribution—yet computational work routinely achieves dimension reduction to low-dimensional aggregates. Can we extend geometric saddle-path analysis to stochastic environments? What does such a framework reveal about equilibrium dynamics, state-dependent responses, and the success of computational approximations?

The paper makes three contributions, all drawn from a single geometric object. First, I define *conditional saddle paths*: equilibrium trajectories under frozen exogenous regimes, traced by the decision rules of the stochastic equilibrium, with agents still pricing future switches. Each regime’s trajectory draws a policy graph in the aggregate phase plane, decomposing fluctuations into movements *along* a conditional saddle (endogenous propagation) and *across* saddles (exogenous shocks)—the stochastic analogue of the deterministic phase diagram. The geometry carries the economics: shock impacts are state-independent when the saddles are parallel; relative tilt makes the same shock hit harder in some states; and even with parallel saddles, the response *shape*—peak timing and persistence—depends on position along the saddle. I develop these in a canonical RBC model (Sections 2–3).

Second, the same geometry speaks to a long-standing computational puzzle: why tracking a single aggregate can stand in for an entire wealth distribution. When an aggregate variable moves monotonically along the conditional saddle, it exactly indexes the equilibrium state within a frozen-regime spell, so one number recovers the distribution and every price. I show that aggregate capital plays this role in [Krusell and Smith \(1998\)](#), near the conditional steady state and, under checkable conditions, on an explicit region away from it. The result is candid about its edge: it must rule out a stall in which capital stops while the distribution rearranges, and across spells a synchronization argument controls the discrepancy once the frozen-regime dynamics contract. A perfect-foresight diagnostic then carries the key condition from a single

deterministic transition path—the kind practitioners already compute—to the full equilibrium (Section 4.1).

Third, the same geometry turns policy comparative statics into a picture. In a search-and-matching economy (Section 4.2), conditional saddles trace regime-specific vacancy loci whose across-regime transitions generate the Beveridge-curve loop, and the surplus compression behind the Shimer puzzle appears as differential saddle steepness. Comparative statics on unemployment benefits yield a new result: volatility is hump-shaped in the benefit level. The reason is geometric—each regime pulls unemployment toward its own conditional steady state, so volatility is governed by the *gap between the conditional steady states*, not slope asymmetry alone; at high benefits the steady states converge despite large slope gaps, leaving little room to fluctuate. Across the three, the conclusions are read from one object: the slope gives state dependence, monotonicity gives dimension reduction, and the spacing of resting points gives comparative statics.

Related literature This paper contributes to three strands of literature in macroeconomics. The first studies global equilibrium dynamics and solution methods under aggregate uncertainty, where characterizing equilibrium dynamics with aggregate shocks has motivated extensive methodological development. [Marcet \(1988\)](#), [Den Haan and Marcet \(1990\)](#), and [Krusell and Smith \(1998\)](#) pioneered bounded rationality approximations; [Krusell and Smith \(1998\)](#) found that a simple linear forecasting rule in aggregate capital achieves remarkable accuracy ($R^2 > 0.9999$) despite the infinite-dimensional state space. Subsequent work sharpened accuracy and efficiency through perturbation and projection methods ([Reiter, 2010](#); [Ahn et al., 2018](#); [Levintal, 2018](#); [Winberry, 2018](#); [Bayer and Luetticke, 2020](#); [Bhandari et al., 2023](#)), simulation and aggregation approaches ([Den Haan, 1996](#); [Den Haan and Rendahl, 2010](#); [Judd et al., 2011](#); [Maliar and Maliar, 2015](#)), adaptive sparse grids ([Brumm and Scheidegger, 2017](#)), sequence-space methods ([Auclert et al., 2021](#); [Azinovic-Yang and Žemlička, 2025](#)), and deep learning ([Maliar et al., 2021](#); [Azinovic et al., 2022](#); [Fernández-Villaverde et al., 2023](#); [Han et al., 2026](#)). Rather than reducing the state, [Childers \(2018\)](#) linearizes in function space, treating the entire distribution as the state, with consistency guarantees. On the theoretical side, [Miao \(2006\)](#) and [Cao \(2020\)](#) establish existence of competitive and recursive equilibria in this class; [Proehl \(2025\)](#) gives conditions for existence and uniqueness under aggregate and idiosyncratic

risk, and [Proehl \(2024\)](#) develops a companion global solution method. The framework here is complementary, taking the recursive equilibrium as given and characterizing the geometry of its dynamics.

The sequence-space Jacobian method of [Auclert et al. \(2021\)](#) reduces dimension by linearizing around a steady state; conditional saddle paths instead operate globally, exploiting monotonicity of an aggregate coordinate along the frozen-regime path. The two coincide to first order near a conditional steady state, but away from it the saddle bends, capturing state-dependent dynamics that first-order methods cannot—such as the regime-dependent slope asymmetries frictions generate (Section 3).

This paper differs from these computational contributions by providing geometric foundations for *why* dimension reduction works: aggregate capital’s within-spell sufficiency follows from nullcline invariance and monotonicity rather than numerical happenstance. This also has computational implications, since along each conditional saddle aggregate capital provides a theoretically justified metric for the repeated transition method of [Lee \(2026a\)](#).¹

Second, this paper extends the geometric tradition of phase-diagram analyses in growth theory ([Solow, 1956](#); [Swan, 1956](#); [Ramsey, 1928](#); [Koopmans, 1963](#); [Cass, 1965](#); [Barro and Sala-i-Martin, 1992](#)) to stochastic environments. Relatedly, the Stable Transformation Path of [Buera et al. \(2024\)](#) extends the balanced growth path, and its turnpike property, to non-stationary growth models, defining a stable path independent of initial conditions. Conditional saddle paths extend the same classical geometry along a different dimension, from deterministic to stochastic settings, providing a visual tool for how microfounded frictions generate state-dependent dynamics, as documented empirically ([Kaplan and Violante, 2014](#); [Vavra, 2014](#); [Berger and Vavra, 2015](#); [Basu and Bundick, 2017](#); [Bloom et al., 2018](#); [Kaplan et al., 2018](#); [Petrosky-Nadeau et al., 2018](#); [Rognlie et al., 2018](#); [Baley and Blanco, 2019](#); [Pizzinelli et al., 2020](#); [Berger et al., 2021](#); [Winberry, 2021](#); [Lee, 2026b](#); [Melcangi, 2024](#)).

Finally, the random dynamical systems literature ([Arnold, 1998](#); [Schenk-Hoppé, 1998, 2001](#)) focuses on history-dependent stochastic saddle paths ([Yannacopoulos,](#)

¹Freezing the aggregate state and analyzing transitions appears in several computational approaches to heterogeneous-agent models, including [Bourany \(2018\)](#), Section 4.3 of a working paper version of [Achdou et al. \(2021\)](#), and [Lee \(2026a\)](#). In continuous time, the master-equation literature ([Achdou et al., 2021](#); [Bilal, 2023](#)) treats the distribution itself as the state and asks when finite-dimensional projections suffice, the rigorous mathematical home of this question. Conditional saddle paths formalize the underlying geometric objects, and the sufficiency results explain when dimension reduction is exact along frozen-regime paths.

2011) that vary with the realized shock sequence.² By contrast, conditional saddle paths are history-invariant geometric objects, the direct analogue of deterministic phase diagrams for economists.

2 Conditional saddle path

2.1 Definitions and assumptions

I consider a generic dynamic stochastic model whose recursive competitive equilibrium (RCE) is characterized by an aggregate state S and laws of motion $(\Gamma_{\text{endo}}, \Gamma_{\text{exo}})$:

$$S = [\Phi, A] \tag{1}$$

where Φ is the endogenous and A the exogenous aggregate state, the latter possibly multivariate. The exogenous law Γ_{exo} is a Markov chain; for the illustration I take a two-state chain with $A \in \{B, G\}$ and $\Gamma_{\text{exo}}(A'|A) > 0 \ \forall A', A$.

The distributional state space $\mathcal{X}$ has elements Φ summarizing the cross-sectional distribution of payoff-relevant idiosyncratic household states (e.g., assets and productivity types) plus any endogenous objects needed to evaluate decision rules; formally $\mathcal{X}$ is a subset of a metric space of probability measures, possibly augmented by a finite-dimensional vector of aggregate variables. Throughout, $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$ maps the current distributional state to next period's under frozen regime A .

The central object formalizes a natural thought experiment: the economy sits at (Φ_0, A) today and the regime stays at A indefinitely. The resulting equilibrium trajectory, computed under decision rules that fully internalize future regime changes, is a frozen-regime orbit, and its closure (visited states plus their limit points) is the conditional saddle path. In aggregate phase diagrams the associated regime-specific locus is a conditional policy graph, to which the slope and curvature statements below refer.

Definition 1 (Frozen-regime orbit and conditional saddle path).

Fix a regime $A \in \{B, G\}$ and an initial state $\Phi_0 \in \mathcal{X}$. Let $\{\Phi_t\}_{t \geq 0}$ denote the frozen-

²In Random Dynamical Systems language, the conditional steady state is a deterministic realization of the random fixed point (Schenk-Hoppé and Schmalfuß, 2001) and the conditional saddle path is the invariant manifold of the frozen-regime dynamics; the synchronization theorem of Section 4.1 (Theorem 2) uses exactly this pullback random-fixed-point construction, recovering the current state as the long-history limit of trajectories started ever further in the past under the same regime history.

regime continuation under A , defined recursively by

$$\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A), \quad t \geq 0. \quad (2)$$

The frozen-regime orbit is

$$\mathcal{O}(\Phi_0, A) := \{\Phi_t : t \geq 0\}. \quad (3)$$

The conditional saddle path under A from Φ_0 is defined as the closure of the frozen-regime orbit:

$$\mathcal{M}(\Phi_0, A) := \overline{\mathcal{O}(\Phi_0, A)}. \quad (4)$$

The space $(\mathcal{X}, d)$ is a metric space: two states are close in d when the averages they assign to any well-behaved function of the individual state are close, finite-dimensional components compared in the Euclidean norm.³ The closure in Definition 1 uses this metric.

Two projection ingredients recur: a *coordinate* $e : \mathcal{X} \rightarrow \mathbb{R}$, a predetermined aggregate variable such as capital K or inherited unemployment u_{-1} ; and an *aggregate object* $g(\Phi, A)$, an equilibrium outcome such as consumption or vacancies.

Definition 2 (Conditional policy graph).

For a regime A , a coordinate e , and an aggregate object g , the conditional policy graph in the (e, g) projection is

$$\mathcal{P}_A^g := \{(e(\Phi), g(\Phi, A)) : \Phi \in \mathcal{D}_A\}, \quad (5)$$

where $\mathcal{D}_A$ is the set of distributional states reached by the frozen- A continuations under study.

When e is injective (one-to-one) on a given conditional saddle path, the restriction of $\mathcal{P}_A^g$ to that path is the graph of a single-valued function of e .

Based on the conditional saddle path, I define the conditional steady state:

Definition 3 (Conditional steady state).

Given (Φ_0, A) , if the frozen-regime continuation $\{\Phi_t\}_{t \geq 0}$ converges, I denote its limit by

$$\Phi^{cs}(\Phi_0, A) := \lim_{t \rightarrow \infty} \Phi_t. \quad (6)$$

³Formally, d is any metric for which $d(\Phi_n, \Phi) \rightarrow 0$ if and only if $\int f d\Phi_n \rightarrow \int f d\Phi$ for every bounded continuous function f ; this is the standard notion of weak convergence.

Fixing A makes the law of motion deterministic on the distribution space: from Φ_0 , $\{\Phi_t\}_{t \geq 0}$ is generated by repeated application of $\Gamma_{\text{endo}}(\cdot, A)$. Thus $\mathcal{M}(\Phi_0, A)$ is the natural analogue of the saddle arm in deterministic saddle-path analysis, isolating the equilibrium states (and their limit points) visited along the convergence dynamics under regime A .⁴

For the conditional saddle to serve as a useful geometric object, a one-dimensional curve traversed during each frozen-regime spell, the dynamics must converge to a unique limit and vary smoothly. The following assumption collects these conditions.

Assumption 1 (Regularity of conditional saddle paths).

Fix a regime $A \in \{B, G\}$ and an initial state $\Phi_0 \in \mathcal{X}$. Let $\{\Phi_t\}_{t \geq 0}$ be defined by $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$.

- (i) *(Unique existence) The conditional steady state $\Phi^{\text{cs}}(\Phi_0, A)$ uniquely exists on $\mathcal{M}(\Phi_0, A)$.*
- (ii) *(Continuity) The map $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$ is continuous. Moreover, the aggregate variables (e.g., aggregate capital K and consumption C) vary continuously in Φ when restricted to $\mathcal{M}(\Phi_0, A)$.*

Part (i) requires that, conditional on regime A and initial state Φ_0 , the frozen-regime continuation converge to a unique conditional steady state; it asserts neither global uniqueness across initial conditions nor the absence of multiplicity across regimes.⁵ Part (ii) imposes continuity of the law of motion and aggregate observables on the conditional saddle, so small perturbations of the state produce small changes in aggregate outcomes. Assumption 1(i) thus restricts attention to convergent continuations, ruling out periodic orbits or divergent paths. In the models below, convergence follows from standard arguments: diminishing returns in the RBC benchmark and contractivity of the Bewley–Huggett–Aiyagari operator (Bewley, 1986; Huggett, 1993; Aiyagari, 1994) in the heterogeneous-household economy.

⁴Strictly, in discrete time $\mathcal{M}(\Phi_0, A) = \overline{\mathcal{O}(\Phi_0, A)}$ is an orbit-closure, not necessarily a smooth manifold (here a smooth curve); saddle-path language emphasizes its one-dimensional, path-like role. Slope and curvature language applies to $\mathcal{P}_A^g$ in a chosen projection; in continuous time the frozen-regime object is a one-dimensional invariant manifold.

⁵Proehl (2025) establishes global existence and uniqueness of the recursive equilibrium under explicit conditions on primitives, whereas Assumption 1(i) allows global multiplicity as in Walsh and Young (2024); conditional on a given initial state and fixed regime, the equilibrium path is assumed uniquely determined.

With this regularity the conditional saddle admits a natural notion of position along the orbit. Theorem 1 (Section 4.1) formalizes the key implication: if some aggregate variable moves strictly one-way along the path, it provides a valid coordinate for the entire conditional saddle, yielding an exact one-dimensional representation on $\mathcal{M}(\Phi_0, A)$.

2.2 Representative-agent economy: A canonical RBC

A representative household with log utility maximizes lifetime utility under stochastic aggregate TFP A_t , given (a_0, A_0) . I use the recursive competitive equilibrium, equivalent to the sequential equilibrium under standard regularity conditions.⁶ The recursive form is:

$$v(a; K, A) = \max_{c, a'} \log(c) + \beta \mathbb{E} v(a'; K', A') \quad (7)$$

$$c + a' = a(1 + r(K, A)) + w(K, A) \quad (8)$$

$$a' \geq -\bar{a} \quad (9)$$

$$K' = \Gamma_{\text{endo}}(K, A), \quad A' \sim \Gamma_{\text{exo}}(A'|A), \quad (10)$$

where v is the value function, a beginning-of-period wealth, K aggregate capital, A aggregate TFP, and Γ_{endo} the law of motion for K . TFP A follows a Markov-switching process between levels B and G , with $G > B$:

$$\Gamma_{\text{exo}} = \begin{bmatrix} \pi_{BB} & \pi_{BG} \\ \pi_{GB} & \pi_{GG} \end{bmatrix} \quad (11)$$

Competitive factor prices from a CRS Cobb–Douglas production function are:

$$r(K, A) = A\alpha K^{\alpha-1} - \delta \quad (12)$$

$$w(K, A) = A(1 - \alpha)K^\alpha, \quad (13)$$

where capital K satisfies $K = a$ in equilibrium. The recursive competitive equilibrium (RCE) is:

Definition 4 (Recursive competitive equilibrium).

$(c, a', v, r, w, \Gamma_{\text{endo}})$ is a recursive competitive equilibrium if:

1. given prices (r, w) , the policies (c, a') attain the value v in the household problem;
2. prices satisfy the competitive factor-price conditions above;

⁶The appendices are collected in a separate Online Appendix.

3. consistency holds:

$$a'(K; K, A) = K' = \Gamma_{\text{endo}}(K, A). \quad (14)$$

Using Γ_{endo} from Definition 4, I define the conditional K -nullcline.

Definition 5 (Conditional K -nullcline in RBC).

Fix a regime $A \in \{B, G\}$. The conditional K -nullcline is the zero-net-investment locus in the (K, C) phase space,

$$\mathcal{N}(A) := \{(K, C) \in \mathbb{R}_{++} \times \mathbb{R}_{++} : \Delta K(K, C; A) = 0\}, \quad (15)$$

where $\Delta K(K, C; A)$ is the one-period change in capital implied by the resource constraint at (K, C) under regime A . When $\mathcal{N}(A)$ is the graph of a function, I write $\mathcal{N}(A) = \{(K, C) : C = C_A^{Knull}(K)\}$.

Along a strictly monotone frozen-regime continuation, the orbit meets $\mathcal{N}(A)$ only in the limit, at the conditional steady state; the locus is thus distinct from the *realized nullcline crossings* $\{(K_t, C_t) : t \geq 0, K_{t+1} = K_t\}$, empty before the limit. The resource constraint $Y = C + I$ with $I = K_{t+1} - (1 - \delta)K_t$ gives $\Delta K = AK^\alpha - \delta K - C$, so $\mathcal{N}(A)$ is an explicit function graph.

Proposition 1 (Characterizing the conditional K -nullcline in RBC).

The conditional nullclines of aggregate capital K for $A \in \{B, G\}$ are

$$C_A^{Knull}(K) = AK^\alpha - \delta K, \quad (16)$$

independent of any initial condition.

Proof. From $\Delta K(K, C; A) = 0$ and $\Delta K = AK^\alpha - \delta K - C$, the nullcline is $C = AK^\alpha - \delta K$ for each A . ■

The nullcline invokes only $\Delta K = AK^\alpha - \delta K - C$, so it bears no imprint of preferences, beliefs, or the distribution; the same holds for the induced *direction field*, whose net-investment sign $\Delta K = C_A^{Knull}(K) - C$ is fixed by accounting alone. Beliefs and heterogeneity relocate the equilibrium orbit within a phase portrait whose boundary and sign structure they cannot move. This is the common root of belief-invariance under aggregate uncertainty (Proposition 3), underlying the perfect-foresight diagnostic of Appendix G, and distribution-invariance in the heterogeneous-household model (Proposition 7), underlying within-spell dimension reduction.

In discrete time capital can jump a finite amount, so the projected (K, C) trajectory can develop folds or loops, pathologies absent in continuous time. I rule them out with a geometric restriction.

Assumption 2 (No segment crossing of the conditional K -nullcline).

For each $A \in \{B, G\}$, along the frozen- A continuation: for every pre-limit transition (every $t \geq 0$ with $K_{t+1} \neq K_t$), the open straight line segment in (K, C) connecting (K_t, C_t) and (K_{t+1}, C_{t+1}) does not intersect the active regime's conditional K -nullcline $\mathcal{N}(A)$:

$$((K_t, C_t), (K_{t+1}, C_{t+1})) \cap \mathcal{N}(A) = \emptyset, \quad \forall t \geq 0 \text{ with } K_{t+1} \neq K_t. \quad (17)$$

Endpoints are not required to lie off the nullcline; in particular, the conditional steady state is allowed to lie on $\mathcal{N}(A)$, and the assumption imposes no condition at the degenerate terminal transition $K_{t+1} = K_t$.

Assumption 2 ensures consecutive points cannot strictly switch sides of the K -nullcline. Its companion is *unique K -stationarity*: $K_{t+1} = K_t$ only at $\Phi_t = \Phi^{cs}(\Phi_0, A)$, which in the representative-agent model follows from Assumption 1(i) (K is Markov under the frozen regime) and is otherwise maintained as a pathwise hypothesis. Economically, Assumption 2 rules out single-period overshooting so severe that capital jumps past its stationary level and reverses, the discrete-time analogue of the continuous-time property that trajectories cannot cross nullclines.⁷

Although stated for the RBC benchmark, Assumption 2 is central to the heterogeneous-household economy (Section 4.1) and can be verified from primitives without solving the full stochastic equilibrium, conditional on a simulated frozen-regime continuation (or certified envelope bounds along it); Appendices G and E develop two complementary strategies.

Proposition 2 (K -monotonicity in RBC).

Fix $A \in \{B, G\}$. Under Assumptions 1 and 2, together with unique K -stationarity ($K_{t+1} = K_t$ only at $\Phi_t = \Phi^{cs}(\Phi_0, A)$ along the frozen- A continuation), aggregate capital converges to K_A^{cs} strictly monotonically.

Proof. Fix A and let $H_t := C_t - C_A^{Knull}(K_t)$ denote the nullcline gap. The resource constraint gives $\Delta K_t = -H_t$. By Assumption 2, no transition strictly switches sides

⁷For related geometric restrictions in dynamic optimization, see Reddy et al. (2020) and, on threshold (Skiba) phenomena, Skiba (1978), Dechert and Nishimura (1983), and Wagoner (2003).

of the nullcline: $H_t H_{t+1} \geq 0$ for every transition t with $K_{t+1} \neq K_t$, and $H_t \neq 0$ on any such transition. Unique K -stationarity confines $H_{t+1} = 0$ to terminal arrival at Φ_A^{cs} , so prior to convergence $\{\Delta K_t\}$ has constant nonzero sign. Unique K -stationarity rules out nonterminal flat steps ($\Delta K_t = 0$ at any $\Phi_t \neq \Phi_A^{cs}$), so every pre-convergence transition is nondegenerate and $\{K_t\}$ is strictly monotone. Convergence $K_t \rightarrow K_A^{cs}$ follows from Assumption 1. $\blacksquare$

Unique K -stationarity as a no-plateau condition Unique K -stationarity rules out *plateau equilibria*: states off the conditional steady state at which aggregate capital stops moving ($\Delta K_t = 0$) while the distribution keeps evolving, excluding the degenerate fixed points of K that arise with poverty traps, occasionally binding constraints, or policy kinks. In the representative-agent RBC, K is Markov under the frozen regime, so the condition reduces to uniqueness of the conditional steady state (Assumption 1(i)); in the heterogeneous-agent economy it holds locally via the stable eigenvalue $\lambda_A^s \in (0, 1)$ of the linearized frozen-regime dynamics (Proposition 10). A globally certified version is more delicate: the bounded price-income MPC of Assumption 3 (Appendix E) constrains the off-steady-state dynamics, but I treat unique K -stationarity as a maintained pathwise condition, checkable along any simulated frozen-regime continuation.

In the representative-agent benchmark these restrictions are not free-standing assumptions but consequences of the scalar concave structure, given two checkable inputs verified in the computed equilibrium: a curvature-margin bound and, for stochastic beliefs, a single-crossing condition on the conditional Euler residual. Since capital is the autonomous scalar state, the frozen- A continuation iterates a single policy $g_A(K)$, and each restriction follows from a primitive:

1. strict concavity of the one-sector problem delivers a unique conditional steady state (under deterministic beliefs; the stochastic case adds the verified single-crossing input, over the capital range on which it is verified);
2. the same concavity delivers strictly monotone convergence to that steady state, with the same qualification;
3. the same concavity, under an explicit curvature-margin bound satisfied with wide slack in standard calibrations, delivers no segment crossing of the nullcline;

4. feasibility bounds capital, so explosion never arises;
5. the transversality condition enters only in selecting the stable policy.

Appendix K gives the proof. The conditions are stated as assumptions because the heterogeneous-agent and search economies of Sections 4.1–4.2 replace this scalar map by an operator on the distribution: K is no longer autonomous, and monotone convergence with a real stable eigenvalue becomes a spectral property of the linearized dynamics on the distribution space that cannot be read off primitives. The non-concave illustration of Appendix E shows the benchmark’s concavity is essential, not decorative.

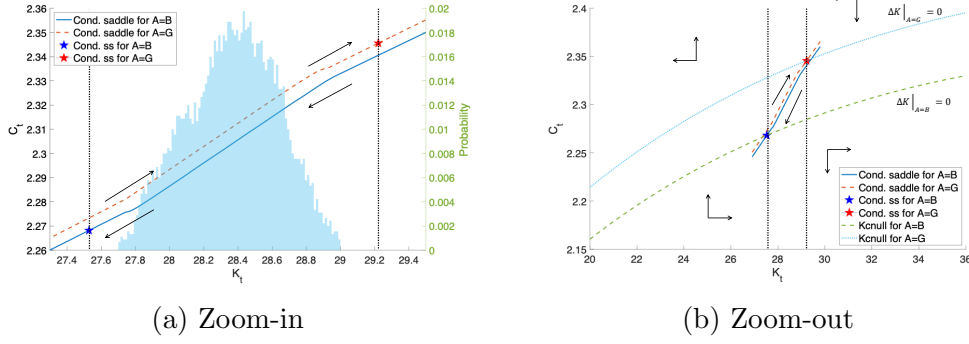
Figure 1 depicts the calibrated RBC model’s conditional saddle paths in the (K, C) plane under $A \in \{B, G\}$, solved globally via the repeated transition method of Lee (2026a).⁸ Panel (a) zooms into the conditional steady states with a histogram of simulated K_t ; panel (b) shows the global partition with conditional K -nullclines and direction arrows. All long-run dynamics stay within the region bounded by the two conditional saddles: within a spell the economy converges along the active saddle, and regime switches move it across, so the entire ergodic distribution is nested within the saddle boundaries.

MIT shock on the saddle This decomposition characterizes unexpected transitory (MIT) shocks, whose link to first-order perturbation in heterogeneous-agent models is established by Boppart et al. (2018). For shocks representable as displacements of the predetermined state under unchanged decision rules, a shift along the conditional saddle replicates the impact and all subsequent dynamics are along-path convergence, so the slope and curvature of the conditional policy graph at the displacement point fully determine propagation, including persistence, amplification, and state dependence.

Generalized transition function Because conditional saddle paths capture all equilibrium transitions, along and across saddles, they support generalized transition function analysis (Lee, 2026a), encompassing generalized impulse response functions (Koop et al., 1996; Andreasen et al., 2017): any shock sequence is traced in the phase

⁸For each regime A , I hold productivity fixed for 2,500 periods and iterate the equilibrium law of motion from initial capital on both sides of K_A^{cs} , recovering the full conditional saddle in the phase diagram.

Figure 1: Conditional saddle paths and the phase diagram



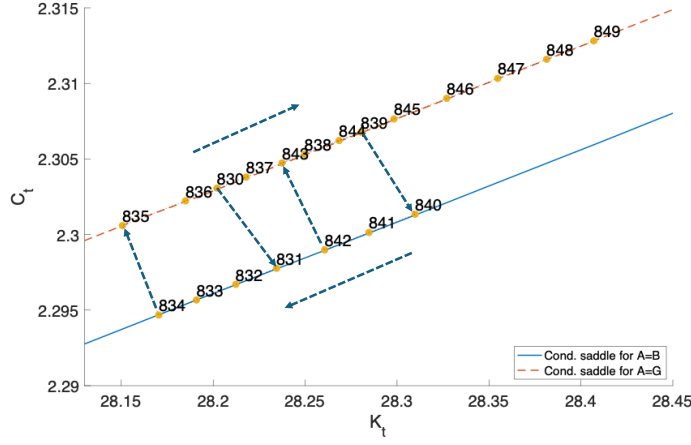
Notes: Conditional K -nullclines ($\Delta K = 0$) partition the (K, C) plane. Direction arrows show frozen-regime convergence along each conditional saddle. Panel (a) zooms into the neighborhood of the conditional steady states shown in panel (b). The histogram plots the time-series distribution of aggregate capital.

diagram as along-saddle and across-saddle movements, not confined to local dynamics around a steady state.

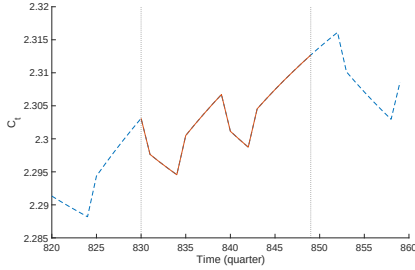
Figure 2 shows equilibrium dynamics in the phase diagram (panel (a)) and time domain (panels (b) and (c)) over a 20-quarter subsample. Negative TFP shocks produce downward jumps in consumption across conditional saddles, followed by endogenous decline along them; positive shocks reverse this. The phase diagram captures the full dynamics in a single figure and permits immediate counterfactual analysis. For the GIRF, I replace the two-state chain with a finely discretized AR(1) for log productivity, $\ln A' = \rho \ln A + \epsilon$, $\epsilon \sim N(0, \sigma^2)$. There is then a continuum of frozen- A conditional saddles indexed by A , and the GIRF path moves both along a given saddle (endogenous propagation) and across the family as A evolves.

Following a positive TFP shock, consumption jumps upward across conditional saddles on impact; thereafter, along- and across-saddle forces jointly generate a bow-shaped trajectory in the phase diagram, with capital hump-shaped as the shock mean-reverts and consumption first rising then falling as TFP crosses the threshold where along-saddle adjustment reverses direction (Figure 2).

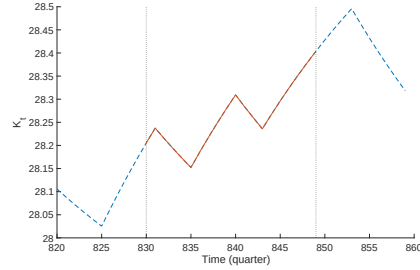
Comparison with the perfect-foresight saddles In the *perfect-foresight* (PF) economy, agents believe the current regime persists forever, so the PF steady states solve $1 = \beta(1 - \delta + \alpha A(K_A^{pf})^{\alpha-1})$. Along the *conditional saddle* (CS), the regime is fixed ex post but agents correctly anticipate switches under Γ_{exo} , so the conditional



(a) On the saddles



(b) C - time domain



(c) K - time domain

Figure 2: Equilibrium dynamics: saddle vs. time domain

Notes: Stochastic equilibrium dynamics on the conditional saddle paths (panel (a)) and in the time domain (panels (b) and (c)). A 20-quarter window from the simulated path (quarters 830–849).

steady states K_A^{cs} are pinned by probability-weighted Euler equations.⁹ The PF and CS economies share the same K -nullcline (Proposition 3) but differ in consumption dynamics.

Proposition 3 (K -nullcline invariance over beliefs).

Conditional K -nullclines are identical between the RBC model with aggregate uncertainty and the perfect-foresight counterpart.

Proof.

From the stationary condition $\delta K = I$ and the aggregate resource constraint $Y = C + I$, equation (16) is immediate. Neither condition involves expectations or belief structures, so the nullcline is identical whether or not there is aggregate uncertainty.

⁹Specifically, $\beta\pi_{AA}(1 + \alpha A(K_A^{cs})^{\alpha-1} - \delta) + \beta\pi_{AA'}(c(K_A^{cs}, A')/c_A^{cs})^{-1}(1 + \alpha A'(K_A^{cs})^{\alpha-1} - \delta) = 1$ for each $A \in \{B, G\}$, where $A' \neq A$.

■

PF–CS differences thus come entirely from consumption dynamics, not the feasibility locus; Proposition 4 formalizes this via steady-state orderings. This invariance is the immovable structural boundary of the framework: an accounting identity indifferent to beliefs and, in Section 4.1, to *who* saves, on which both the dimension-reduction result and the perfect-foresight diagnostic rest.

Proposition 4 (Aggregate uncertainty and the conditional steady states).

Let $R(A, K) := 1 - \delta + \alpha AK^{\alpha-1}$ denote the gross return and $c(K, A')$ the stochastic-equilibrium consumption function. Suppose that (i) at each regime’s PF steady state, the consumption response to a regime switch dominates the return response (jump dominance),

$$\frac{c(K_B^{pf}, G)}{c_B^{pf}} > \frac{R(G, K_B^{pf})}{R(B, K_B^{pf})}, \quad \frac{c(K_G^{pf}, B)}{c_G^{pf}} < \frac{R(B, K_G^{pf})}{R(G, K_G^{pf})}; \quad (18)$$

and (ii) each conditional Euler residual F_A , the residual of the conditional steady-state Euler equation viewed as a function of K (Appendix A), is strictly positive below K_A^{cs} and strictly negative above it. Then

$$K_B^{cs} < K_B^{pf} < K_G^{pf} < K_G^{cs}, \quad c_B^{cs} < c_B^{pf} < c_G^{pf} < c_G^{cs}. \quad (19)$$

Proof.

See Appendix A. ■

Both hypotheses are explicit and directly checkable: a persistent regime switch barely moves the gross return (only the marginal-product term shifts) while consumption moves roughly one-for-one, so condition (i) fails only in degenerate cases. Under full depreciation ($\delta = 1$) the decision rules collapse to the belief-independent Brock–Mirman form, condition (i) holds with equality, and the two sets of steady states coincide (Appendix A); away from that corner the orderings hold in the computed equilibrium.

PF steady states are nested within the CS steady states: hedging against regime switches pushes the conditional steady states apart (Deaton, 1991; Carroll, 1997). Conversely, the conditional saddle *paths* are nested within the PF paths, since at any K hedging compresses the cross-regime consumption gap. Both nestings share one mechanism: compressed consumption responses cumulate into amplified long-run capital divergence.

Conditional boundary condition The recursive formulation imposes a stochastic analogue of the transversality condition. Its conditional counterpart, $\lim_{t \rightarrow \infty} \beta^t u'(c_t) K_t = 0$ along each frozen-regime saddle, rules out explosive continuations and, under standard assumptions, is equivalent to the usual stochastic transversality conditions (Kamihigashi, 2003, 2005), selecting the same bounded RCE regime-by-regime on $\mathcal{M}(\Phi_0, A)$.

Economies without conditional saddle paths A conditional saddle requires a predetermined endogenous state. Without one, the economy jumps each period to the allocation implied by the current exogenous state, so the frozen- A conditional “saddle” collapses to a singleton; I call such an economy *endogenously stateless*.¹⁰ Models with predetermined states (capital, habits, interest-rate smoothing, or distributional states in HANK) are endogenously stateful, and conditional saddle paths exist under mild regularity conditions.

3 State dependence

I analyze nonlinear shock responsiveness through conditional saddle paths. In any stochastic dynamic model admitting such paths, an aggregate variable’s response to a shock is a shift across conditional policy graphs. The benchmark is *state independence*: the impact gap between graphs is constant in the endogenous coordinate; departures show up locally as slope differences, the first-order diagnostic of state-dependent responses. Empirically, responses to the same impulse vary with the business-cycle phase: unemployment dynamics are asymmetric (Neftci, 1984), fiscal multipliers are larger in recessions (Auerbach and Gorodnichenko, 2012), and monetary policy is weaker in recessions (Tenreyro and Thwaites, 2016).

Lemma 1 (Vertical-separation benchmark).

Let I be an interval over which the regime-specific conditional policy graphs $\mathcal{P}_{A_j}^g = \{(e, g_j(e)) : e \in I\}$, $j \in \{0, 1\}$, are single-valued. Define the impact gap between the two graphs as $\Delta_g(e; A_1, A_0) := g_1(e) - g_0(e)$. Then conditions (i) and (ii) below are equivalent; if g_0, g_1 are differentiable on I , all three are equivalent:

(i) Δ_g is constant on I ;

¹⁰The textbook three-equation New Keynesian model (Gali, 2008) under determinacy is endogenously stateless: $(x_t, \pi_t, i_t) = \Psi s_t$ for exogenous state $s_t := (r_t^n, u_t, \nu_t)$, so freezing s_t produces no nontrivial dynamics.

(ii) $\mathcal{P}_{A_1}^g$ is a vertical translation of $\mathcal{P}_{A_0}^g$;

(iii) the slope gap $g'_1 - g'_0$ vanishes on I .

Proof. The first equivalence is definitional; the second is the mean value theorem applied to Δ_g . $\blacksquare$

Lemma 1 is the benchmark; the substantive claim, that the slope gap reflects primitives, is Proposition 5. State independence is a knife-edge: differential tilt across conditional policy graphs, whether from real (Winberry, 2021; Lee, 2026a), financial (Melcangi, 2024), or labor market frictions (Petrosky-Nadeau et al., 2018; Pizzinelli et al., 2020), generates state-dependent shock responses. In the log-linearized frozen-regime RBC dynamics near K_A^{cs} , let $R_A^{cs} := 1 + f'(K_A^{cs}; A) - \delta$ denote the gross return at the conditional steady state, $\lambda_A^s \in (0, 1)$ the stable eigenvalue, and $\psi_A := dC/dK|_{\text{saddle}, A}$ the level saddle slope at K_A^{cs} .

Proposition 5 (Slope ratio as a structural diagnostic).

The level saddle slope admits the closed form

$$\psi_A = R_A^{cs} - \lambda_A^s, \quad (20)$$

and hence the cross-regime slope ratio is

$$\mathcal{R} := \frac{\psi_B}{\psi_G} = \frac{R_B^{cs} - \lambda_B^s}{R_G^{cs} - \lambda_G^s}. \quad (21)$$

Proof. Log-linearize the frozen-regime budget constraint $K_{t+1} = f(K_t; A) + (1 - \delta)K_t - C_t$ around (K_A^{cs}, C_A^{cs}) :

$$\hat{k}_{t+1} = R_A^{cs} \hat{k}_t - \frac{C_A^{cs}}{K_A^{cs}} \hat{c}_t. \quad (22)$$

On the stable manifold (the saddle-path branch of the linearized dynamics), $\hat{c}_t = \psi_A^{\log} \hat{k}_t$ for some log slope $\psi_A^{\log}$. Substituting into (22) yields $\hat{k}_{t+1} = [R_A^{cs} - \frac{C_A^{cs}}{K_A^{cs}} \psi_A^{\log}] \hat{k}_t$, which by definition equals $\lambda_A^s \hat{k}_t$. Solving, $\psi_A^{\log} = (K_A^{cs}/C_A^{cs})(R_A^{cs} - \lambda_A^s)$. Converting to level units via $dC/dK|_{K^{cs}} = (C_A^{cs}/K_A^{cs}) \psi_A^{\log}$ gives (20), and (21) follows by taking the ratio.

Under perfect-foresight expectations, the steady-state Euler condition gives $R_A^{cs} = 1/\beta$ and the formula reduces to $\psi_A = 1/\beta - \lambda_A^s$. Under stochastic expectations over regime transitions, R_A^{cs} is pinned down by $1 = \beta \mathbb{E}_A[(C_A^{cs}/C(\Phi_A^{cs}, A'))^\sigma (1 + f'(K_A^{cs}; A') - \delta)]$ (the return evaluated at tomorrow's capital K_A^{cs} under the realized regime A') and generally differs from $1/\beta$; the formula remains valid with R_A^{cs} evaluated at the (stochastic) conditional steady state. $\blacksquare$

Proposition 5 gives $\mathcal{R}$ a structural interpretation: each regime’s saddle slope is the gap between the gross return on capital at its conditional steady state and the stable eigenvalue, both computable from a local linearization of the frozen-regime dynamics without solving the global policy. One reads off (R_A^{cs}, λ_A^s) and applies (21). Local departures from $\mathcal{R} = 1$ are the geometric signature of state-dependent impact responses, and $\mathcal{R}$ is a model-implied moment identifiable in principle from state-dependent local projections (Auerbach and Gorodnichenko, 2012; Tenreyro and Thwaites, 2016).

The conditional saddle path is a discrete orbit, not a smooth curve; Definition 1 introduces $\mathcal{M}(\Phi_0, A)$ as an orbit closure (generically a countable orbit with its limit point). The slope, curvature, and “tangent” language of Propositions 5 and 6, of Section 4.1, and of Proposition 10 refers to the local C^1 parametrization of $\mathcal{M}(\Phi_0, A)$ near K_A^{cs} , which exists under the regularity conditions of Proposition 10 (Appendix D). Outside this neighborhood, the orbit is the discrete object on which Theorem 1 guarantees injectivity: it never revisits a capital level.

A minimal example: asymmetric adjustment cost I add $\mathcal{C}(K', K) = \frac{\tilde{\mu}}{2}((K' - K)/K)^2 K$, where $\tilde{\mu} = \mu_+$ if $K' - K \geq 0$ and $\tilde{\mu} = \mu_-$ otherwise. The curvature therefore changes with the sign of net capital accumulation, not gross investment $I = K' - (1 - \delta)K$. The specification is C^1 at $K' = K$ but has different second derivatives on its two sides. I interpret $\mu_+ > \mu_-$ as a reduced-form expansion bottleneck—installation, ramp-up, or congestion—rather than a resale wedge.¹¹

The threshold and orientation differ from costly reversibility: Abel and Eberly (1996) formalize it as a purchase price above the resale price, Cooper and Haltiwanger (2006) estimate a mixture of convex, nonconvex, and resale-price costs, and Zhang (2005) uses asymmetric quadratic curvature with $\theta^- > \theta^+$ switching at gross investment $I = 0$. Because $K' = I + (1 - \delta)K$, Zhang’s threshold is $K' = (1 - \delta)K$; mine is $K' = K$, putting high curvature on the opposite side. The specification is a parsimonious laboratory for how an asymmetric real friction tilts policy graphs.

¹¹The calibration maintains the arithmetic-mean curvature $\bar{\mu} = (\mu_+ + \mu_-)/2 = 3.05$ and sets $r = \mu_+/\mu_- = 3$, hence $(\mu_-, \mu_+) = (1.53, 4.57)$. The remaining annual parameters are $\beta = 0.96$, $\delta = 0.10$, and $\alpha = 0.33$; $A \in \{0.98, 1.02\}$ has transition matrix with diagonal entries 0.90. Appendix H details the calibration: the ratio is disciplined by an outlier-robust asymmetry moment (the downside/upside semideviation ratio of investment growth) rather than the raw skewness, and $\mathcal{R}$ is reported across the robustness band $r \in [2, 4]$. This is a conditional discipline on the asymmetry plus an external anchor on the level, not a structural estimate of the two curvature parameters.

All same-capital comparisons use interior common-support domains. Within a given specification, define $\mathcal{R}(K) = C'_B(K)/C'_G(K)$ at the same K over the intersection of the regimes' 5th–95th percentile policy supports. Under the calibrated $r = 3$ specification, the median is 1.265. Holding $\bar{\mu} = 3.05$ fixed, equal curvature gives 0.976, while reversing the pair to $(\mu_-, \mu_+) = (4.57, 1.53)$ gives 0.753. Thus reversing the friction reverses the saddle fan. I drop the earlier estimate 1.83 because neither regime-specific fixed point lies inside the simulated support: clamped extrapolation, rather than an interior policy derivative, generated that value.

Figure 3 examines whether this policy tilt is robust to preferences. Every case keeps $r = 3$, equivalently $(\mu_-, \mu_+) = (1.53, 4.57)$ at $\bar{\mu} = 3.05$, together with the same technology, shock process, and simulated shock path. The left column reports consumption saddles $C_A(K)$; the right reports marginal $q_A^K(K) = 1 + \mathcal{C}_{K'} = 1 + \tilde{\mu}(K' - K)/K$. The top row varies the CRRA coefficient γ , changing relative risk aversion and the EIS, $1/\gamma$, jointly. As γ rises from 1 to 10, the median consumption-slope ratio C'_B/C'_G falls from 1.265 to 1.047. The median absolute marginal- q slope gap $|q_G^{K'} - q_B^{K'}|$ falls from 0.0577 to 0.0307 as both saddles flatten, but the conditional level gap $\mathbb{E}[q^K | G] - \mathbb{E}[q^K | B]$ rises from 0.0153 to 0.0315 and the simulated unconditional $\text{sd}(q^K)$ rises from 0.00932 to 0.01723. Marginal q therefore becomes less sensitive to capital position within a regime but more volatile because regime switches traverse more widely separated saddles.

The bottom row provides the controlled preference comparison. Epstein–Zin utility holds the EIS fixed at one while relative risk aversion rises from 1 to 10. The curves barely move: the median consumption-slope ratio remains approximately 1.265, the absolute marginal- q slope gap changes only from 0.0577 to 0.0576, and $\text{sd}(q^K)$ changes only from 0.00932 to 0.00933. The strong CRRA comparative statics therefore come predominantly from the accompanying fall in the EIS, not from higher risk aversion by itself.

The friction parameters come from the empirical calibration of Appendix H. Holding $\bar{\mu} = 3.05$, I choose r to match the asymmetry of annual investment growth (PNFICA) over 1967–2019, targeting the outlier-robust downside/upside semideviation ratio because the third-moment skewness is fragile to the 2009 outlier. This yields $r = 3$, hence $(\mu_-, \mu_+) = (1.53, 4.57)$; it is a conditional discipline on the asymmetry, not a structural estimate, and $\mathcal{R}$ is robust across $r \in [2, 4]$.

Relative investment volatility is untargeted: the $\pm 2\%$ two-state TFP process caps

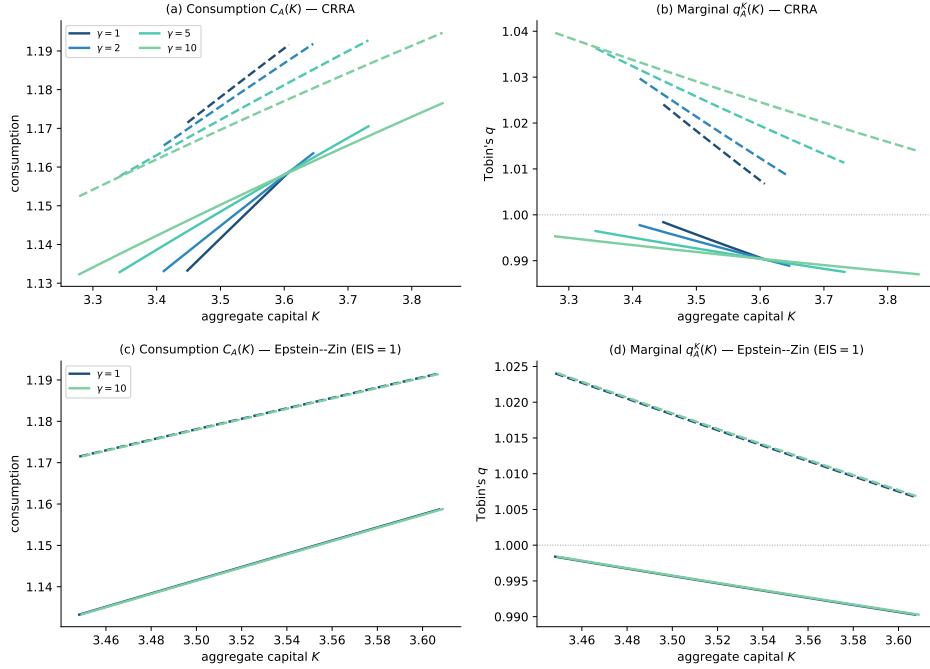


Figure 3: Consumption and marginal- q saddles across preferences

Notes: Panels (a) and (c) plot the bad- and good-regime consumption policy $C_A(K)$; panels (b) and (d) plot marginal $q_A^K(K)$. Colors and markers denote relative risk aversion; solid and dashed lines denote the bad and good regimes. Panels (a) and (b) vary relative risk aversion under CRRA, which simultaneously changes the EIS because $EIS = 1/\gamma$. Panels (c) and (d) use Epstein–Zin utility to vary relative risk aversion while holding the EIS fixed at one; their near overlap shows that risk aversion alone has little effect on these saddles in this calibration. Every panel retains $r = 3$, equivalently $(\mu_-, \mu_+) = (1.53, 4.57)$, a 1:3 contraction-to-expansion curvature ratio at $\bar{\mu} = 3.05$. Every curve is evaluated at the same K on the intersection of the 5th–95th percentile solved supports across all regime–preference cases. Horizontal lines in the marginal- q panels mark $q^K = 1$, or zero net capital accumulation. Marginal q directly displays the assumed adjustment wedge and is not independent calibration evidence. Same-mean symmetric and sign-reversed cases remain in Table 1 and the replication outputs.

it below the data even at negligible friction (Appendix H), so the level is anchored on the investment- q elasticity instead. At the same mean curvature, the symmetric and sign-reversed specifications give median slope ratios of 0.976 and 0.753 versus 1.265 under the target, so the friction’s orientation drives the saddle fan.

The direct external motivation is state-dependent investment responsiveness: [Bachmann et al. \(2013\)](#) document stronger aggregate investment responsiveness in expansions, while [Winberry \(2021\)](#) generates procyclical responsiveness through the extensive margin. These findings motivate an untargeted qualitative comparison: $\mathcal{R}(K)$ is the paper’s model-side geometric diagnostic, not a directly comparable empirical elasticity. By Lemma 1, state-independent impacts correspond to constant vertical

Table 1: Asymmetric adjustment costs: robust-asymmetry calibration and same-capital slopes

Case	μ_-	μ_+	$\sigma(I^{cyc})/\sigma(Y^{cyc})$	σ_-/σ_+	skew($\Delta \log I$)	med $\mathcal{R}(K)$	$\mathcal{C}/Y$
Data	–	–	3.21	1.32	-1.03	–	–
Targeted ($r = 3$)	1.53	4.57	1.85	1.14	-0.63	1.265	0.0032%
Same-mean symmetric ($r = 1$)	3.05	3.05	1.79	1.01	-0.07	0.976	0.0033%
Same-mean sign reversal ($r = 1/3$)	4.57	1.53	1.81	0.90	0.51	0.753	0.0032%

Notes: The asymmetry ratio $r = 3$ is disciplined by the outlier-robust downside/upside semideviation ratio of $\Delta \log(\text{PNFICA})$ over 1967–2019 (Appendix H); the moment skewness is reported for reference only, as it is fragile to the 2009 outlier. I^{cyc} and Y^{cyc} are annual HP-filtered log investment and output with $\lambda = 6.25$; their relative volatility is untargeted and cannot identify the level. $\mathcal{R}(K) = C'_B(K)/C'_G(K)$ is summarized by its median on the common interior capital support. $\mathcal{C}/Y$ is the mean resource cost.

separation; differential tilt reveals the endogenous-state dependence generated by the friction.

Having used the asymmetric friction to isolate slope-driven state dependence, I now return to the frictionless RBC benchmark to study a distinct channel: variation in response shape generated by initial position along the saddle.

Response shape in the frictionless benchmark Although $\mathcal{R} \approx 1$ in this benchmark, the *shape* of the response trajectory is highly state dependent. A positive productivity shock displaces the economy across saddles, and the trajectory depends on K_0 's position relative to K_G^{cs} . Far below K_G^{cs} , the marginal product of capital is high, tilting the intertemporal tradeoff toward accumulation and producing a pronounced, late consumption hump (Barro and Sala-i-Martin, 1992); near K_G^{cs} , across-saddle reversion dominates quickly and the peak arrives earlier. This channel is orthogonal to the slope asymmetry of Proposition 5: slope differences govern *impact* magnitudes (cross-saddle dimension), position-dependent convergence speed governs response *shapes* (along-saddle dimension). The shape does *not* require a nonlinear saddle: even in the log-linearized economy, where the conditional saddle is a straight line, peak timing varies with K_0 . The mechanism is a race between the endogenous along-saddle rate λ and the exogenous across-saddle rate ρ , weighted by initial distance from steady state; concavity of $f(K)$ amplifies the effect but is not its source.

Near the non-conditional steady state, the Blanchard–Kahn decomposition (Blan-

chard and Kahn, 1980) on the saddle branch yields

$$\hat{c}_t = \underbrace{\mathcal{P}(\hat{k}_0)}_{\text{along saddle}} \lambda^t + \underbrace{\mathcal{Q}}_{\text{across saddle}} \rho^t, \quad (23)$$

where $\hat{c}_t$ and $\hat{k}_0$ are log-deviations from steady state, $\lambda \in (0, 1)$ is the stable eigenvalue of the capital dynamics, and $\rho \in (0, 1)$ is the TFP persistence. The shock size $\varepsilon > 0$ enters only through $\mathcal{P} = \psi_k[\hat{k}_0 + \theta\varepsilon/(\lambda - \rho)]$, which depends on initial capital via the saddle slope $\psi_k > 0$ and the capital-TFP elasticity $\theta > 0$, and $\mathcal{Q} = [\psi_a - \psi_k\theta/(\lambda - \rho)]\varepsilon$, which does not.¹² The hump arises from the tension between two decaying exponentials: the λ^t term captures along-saddle propagation, the ρ^t term direct TFP reversion, and the peak occurs where they balance. Lower initial capital reduces $\mathcal{P}(\hat{k}_0)$, weakening the slowly decaying along-saddle term, so the balance tips later: exactly the comparative static $\partial t^*/\partial \hat{k}_0 < 0$ below.

Proposition 6 (State-dependent peak timing).

Suppose $\lambda > \rho$, $\mathcal{P}(\hat{k}_0) > 0$, and the consumption response (23) admits a peak at $t^* > 0$. Then

$$\frac{\partial t^*}{\partial \hat{k}_0} = \frac{-\psi_k}{\mathcal{P}(\hat{k}_0) \cdot \ln(\lambda/\rho)} < 0, \quad (24)$$

where $\hat{k}_0$ is the log-deviation of the initial capital stock from steady state.

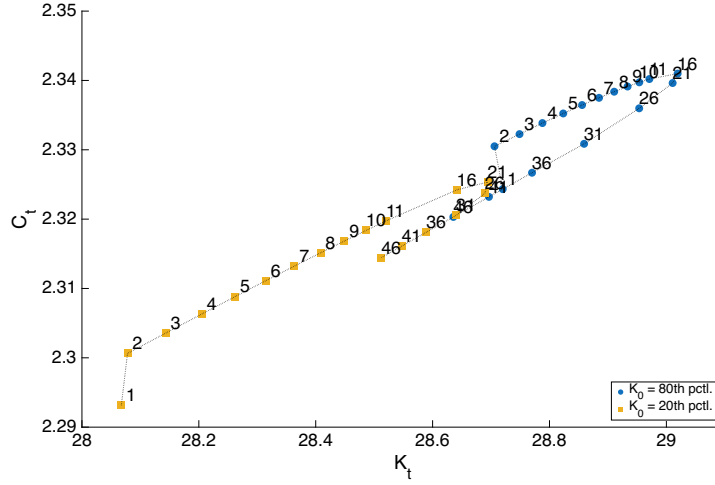
Proof.

See Appendix B. ■

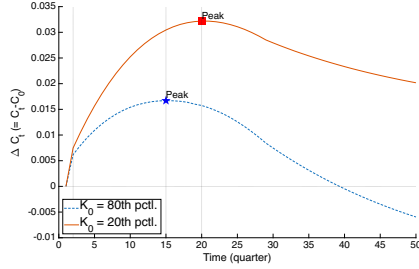
The consumption peak arrives strictly later when initial capital is farther below the steady state. The hypotheses are mild: $\lambda > \rho$ holds under standard quarterly RBC calibrations ($\lambda \approx 0.96$, $\rho \leq 0.95$), and $\mathcal{P}(\hat{k}_0) > 0$ requires only $\hat{k}_0 > -\theta\varepsilon/(\lambda - \rho)$, which the ergodic distribution satisfies with substantial margin. Proposition 6 is local by construction; Figure 4 confirms the mechanism operates globally across the ergodic capital distribution.

Comparing the 20th and 80th percentiles of the ergodic capital distribution: from low K_0 , both C and K rise longer before reversion dominates, producing a later, larger consumption peak; from high K_0 , the peak arrives earlier. The same saddle geometry, shock, and preferences apply, and only the inherited capital stock differs.

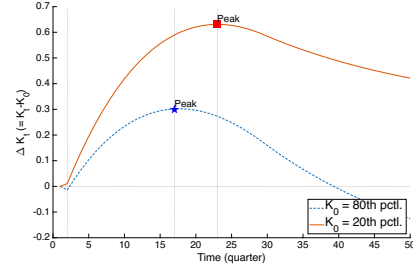
¹²This λ (at the non-conditional steady state) and the regime-specific λ_A^s of Proposition 5 (at the conditional steady state) coincide to first order in the frictionless RBC ($\lambda_B^s = \lambda_G^s = \lambda$, so $\mathcal{R} \approx 1$);



(a) On the saddles



(b) C : time domain



(c) K : time domain

Figure 4: Shape of the impulse responses: low K_0 vs. high K_0

Notes: Impulse responses to a positive productivity shock from K_0 at the 20th percentile and 80th percentile of the ergodic distribution. Panel (a): post-shock trajectories in (K, C) space on the conditional saddle paths; numbers indicate quarters since impact. Panels (b) and (c): time-domain responses of consumption and capital; vertical markers indicate the peak quarter. Sample: 50 quarters.

4 Applications

The conditional-saddle framework applies across model classes, and the two applications below each exercise a different geometric property. In the heterogeneous-household economy (Section 4.1), the *monotonicity* of the saddle delivers dimension reduction; in the search-and-matching economy (Section 4.2), the *spacing* of its resting points delivers comparative statics. The spacing principle already surfaced in the asymmetric-adjustment-cost example of Section 3, where preference-driven separation

frictions generate $\mathcal{R} \neq 1$. Proposition 6 then operates in the $\mathcal{R} \approx 1$ regime, its state dependence driven by initial location $\hat{k}_0$ rather than regime-varying slopes.

of the price saddles governed asset-price volatility; it reappears here as policy-driven separation of the conditional steady states governing unemployment volatility.

4.1 Heterogeneous households: dimension reduction

In a heterogeneous-household economy the natural state is the entire cross-sectional wealth distribution, an infinite-dimensional object. Classical aggregation (Gorman, 1961) reduces dimensionality by restricting preferences, but these restrictions fail under uninsurable idiosyncratic risk and borrowing constraints. Yet practitioners routinely approximate the state by a single number, aggregate capital. The conditional saddle framework explains why this works without preference restrictions by splitting the claim into two halves of very different difficulty: *within* a frozen-regime spell the distribution evolves along a one-dimensional orbit, so aggregate capital is an *exact* coordinate for it; *across* histories—where the classical puzzle actually resides—a single capital rule works only approximately, an error this section bounds and documents directly.

I consider a continuum of ex-ante homogeneous households of unit measure, with recursive problem

$$v(a, z; \Phi, A) = \max_{c, a'} \log(c) + \beta \mathbb{E}v(a', z'; \Phi', A') \quad (25)$$

$$c + a' = a(1 + r(\Phi, A)) + w(\Phi, A)z \quad (26)$$

$$a' \geq 0 \quad (27)$$

$$\Phi' = \Gamma_{\text{endo}}(\Phi, A), \quad A' \sim \Gamma_{\text{exo}}(A'|A). \quad (28)$$

The problem differs from the representative-household economy only in (i) uninsurable idiosyncratic labor productivity, a Markov process $z \sim \Gamma_z(z'|z)$; (ii) inclusion of the distribution Φ in the aggregate endogenous state; and (iii) the corresponding law of motion. The RCE is defined as in Krusell and Smith (1998).

The model has two exogenous stochastic processes, idiosyncratic productivity and aggregate TFP, hence two layers of conditional saddle paths, individual and aggregate. I focus on the aggregate path under stochastic TFP, following Krusell and Smith (1998).¹³ Consistent with Section 2.2, $K = K(\Phi)$ and $C = C(\Phi, A)$ denote aggregate capital and consumption.

¹³In the original Krusell and Smith (1998) model, exogenous individual labor supply co-moves with aggregate TFP. All results are unaffected, but for expositional brevity I assume labor supply is exogenously fixed.

The conditional K -nullcline of Definition 5 extends directly to the heterogeneous-household setting: net investment depends on Φ only through the aggregates (K, C) , so the locus remains an object in the aggregate phase plane.

Definition 6 (Conditional K -nullcline in the heterogeneous-household setting).

Fix $A \in \{B, G\}$. The conditional K -nullcline is the zero-net-investment locus in the aggregate (K, C) plane,

$$\mathcal{N}(A) := \{(K, C) \in \mathbb{R}_{++} \times \mathbb{R}_{++} : \Delta K(K, C; A) = 0\},$$

where $\Delta K(K, C; A)$ is the one-period change in aggregate capital implied by the aggregate resource constraint at (K, C) under regime A . As in the representative-agent case, $\mathcal{N}(A)$ is distinct from the realized crossings $\{(K_t, C_t) : K_{t+1} = K_t\}$ along any frozen-regime continuation $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$, which under strict K -monotonicity contain no point at all before the limit.

Krusell and Smith (1998) posits an endogenous law of motion tracking aggregate capital K rather than the full distribution Φ : $\log K' = b_0(A) + b_1(A) \log K$. This embeds two distinct assumptions: (i) aggregate capital K suffices to summarize the endogenous aggregate state, and (ii) K follows a log-linear law of motion.

Since Krusell and Smith (1998) it has been an open question whether K -based forecasting works so well ($R^2 > 0.9999$) by accident or through a deeper structural property.¹⁴ The conditional-saddle framework splits (i) into two questions of very different difficulty. *Within* a frozen-regime spell, K exactly indexes the distributional state on the conditional saddle, so conditioning on K loses no within-spell information; because the coordinate is strictly monotone along a single orbit, this holds path by path. The economically substantive question is the across-history one at the heart of the Krusell and Smith (1998) puzzle: why approximately the *same* rule $K' = f(K, A)$ works from every distributional state, not merely along one orbit. Proposition 9 bounds the across-spell discrepancy under a uniform contraction condition: the discrepancy decays with shared regime history and vanishes as regime persistence grows. Two caveats keep the claim honest: the contraction condition is maintained rather than derived from primitives, and the bound is loose in the standard calibration. The near-exact accuracy is instead documented directly, by comparing

¹⁴Den Haan (2010) shows the forecasting R^2 can substantially overstate dynamic accuracy, because one-step-ahead fit masks the accumulation of errors over long simulated paths; the across-spell handoff check reported below is in the spirit of his stricter multi-period standard.

the globally solved K' against the (K, A) -based prediction at spell boundaries (an average discrepancy of 2.5×10^{-4} in $\log K$). The synchronization theorem identifies the mechanism that *would* make one rule accurate across histories; the direct check confirms the accuracy is present in this calibration. The same geometry indicates where approximate aggregation should strain: when nonconvexities flatten the null-cline gap or generate multiple K -stationary points, as in lumpy-investment economies where the cross-sectional distribution carries independent forecasting power (Khan and Thomas, 2008; Bachmann et al., 2013), the monotonicity hypothesis weakens and the framework predicts degraded scalar-state accuracy rather than assuming it away.

By contrast, (ii) remains a parametric approximation—the true law of motion for K is generally non-Markovian and need not be log-linear—a matter of functional form, not coordinate choice. Within a spell richer functional forms sharpen accuracy; across spells additional moments can help, with gain bounded by the across-spell discrepancy, small in the calibration examined here.

The following theorem is the first step toward exact conditional sufficiency of K : if the economy converges along a one-dimensional equilibrium path and some aggregate variable moves strictly in one direction throughout, its current value tells you exactly where the economy is, since no two distinct distributional states share the same value.

Theorem 1 (Monotone aggregate variable as a coordinate).

Fix (Φ_0, A_0) and $A \in \{B, G\}$. Let $\{\Phi_t\}_{t \geq 0}$ be the frozen-regime continuation $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$ and let $\Phi^{cs} := \lim_{t \rightarrow \infty} \Phi_t$. Let $e : \mathcal{M}(\Phi_0, A) \rightarrow \mathbb{R}$ be an aggregate equilibrium variable. Suppose that the scalar sequence $\{e(\Phi_t)\}_{t \geq 0}$ is strictly monotone and converges to $e(\Phi^{cs})$. Then e uniquely indexes states on the conditional saddle:

$$e(\Phi) = e(\Phi') \implies \Phi = \Phi' \quad \forall \Phi, \Phi' \in \mathcal{M}(\Phi_0, A). \quad (29)$$

Proof. Define $\iota_A : \mathbb{N} \cup \{\infty\} \rightarrow \mathcal{M}(\Phi_0, A)$ by $\iota_A(t) := \Phi_t$ for $t \in \mathbb{N}$ and $\iota_A(\infty) := \Phi^{cs}$. By definition of $\mathcal{M}(\Phi_0, A)$ and $\Phi_t \rightarrow \Phi^{cs}$,

$$\mathcal{M}(\Phi_0, A) = \{\Phi_t : t \geq 0\} \cup \{\Phi^{cs}\}, \quad (30)$$

so ι_A is surjective. Strict monotonicity of $\{e(\Phi_t)\}$ implies $e(\Phi_t) \neq e(\Phi_\tau)$ for $t \neq \tau$, hence $\Phi_t \neq \Phi_\tau$, so ι_A is injective on $\mathbb{N}$; convergence to $e(\Phi^{cs})$ gives $e(\Phi_t) \neq e(\Phi^{cs})$ for all finite t , so ι_A is injective on $\mathbb{N} \cup \{\infty\}$, hence a bijection. For $\Phi = \iota_A(s)$, $\Phi' = \iota_A(s')$, $e(\Phi) = e(\Phi')$ gives $e(\iota_A(s)) = e(\iota_A(s'))$; since $e \circ \iota_A$ is injective, $s = s'$ and $\Phi = \Phi'$. Thus e is injective on $\mathcal{M}(\Phi_0, A)$, and an inverse map φ_A exists on $e(\mathcal{M}(\Phi_0, A))$. ■

Consequently, a unique inverse map $\varphi_A : e(\mathcal{M}(\Phi_0, A)) \rightarrow \mathcal{M}(\Phi_0, A)$ satisfies $\varphi_A(e(\Phi)) = \Phi$ on $\mathcal{M}(\Phi_0, A)$, and any equilibrium object restricted to $\mathcal{M}(\Phi_0, A)$ is a single-valued function of the scalar coordinate e .

The following proposition shows that the conditional K -nullclines coincide with those of the representative-agent model.

Proposition 7 (Conditional K -nullclines of Krusell and Smith (1998)).

The conditional K -nullclines of the heterogeneous-household model are identical to those of the representative-household model and invariant over the initial distribution Φ_0 :

$$C_A^{Knull}(K) = AK^\alpha - \delta K, \text{ for } A \in \{B, G\}. \quad (31)$$

Proof.

As in Proposition 3, the stationary condition $\delta K = I$ and the aggregate resource constraint $Y = C + I$ immediately imply the form of the conditional nullclines. ■

As anticipated after Proposition 1, this is the distribution-invariance instance of that common root, and it is what enables dimension reduction: the boundary governing the sign of capital accumulation, an accounting identity indifferent to *who* saves, does not move as the distribution evolves along the conditional saddle.

The invariance is specific to the neoclassical production structure with exogenous labor supply. Under separable CRRA preferences with wealth effects on labor, labor supply introduces a distributional channel into the resource constraint; the nullcline then acquires distributional dependence, blocking the analytical route to K -sufficiency (though the perfect-foresight diagnostic of Appendix G remains applicable).

Proposition 8 (K monotonicity and injectivity).

Fix $A \in \{B, G\}$. Under Assumptions 1 and 2, together with unique K -stationarity along the frozen- A continuation (the no-plateau condition of Section 2.2), aggregate capital converges to K_A^{cs} strictly monotonically on $\mathcal{M}(\Phi_0, A)$. Consequently, K is injective on $\mathcal{M}(\Phi_0, A)$.

Proof.

By Proposition 7, the conditional K -nullcline takes the RBC form, so strict monotonicity follows the proof of Proposition 2 verbatim with $C_A^{Knull}(K) = AK^\alpha - \delta K$. Unique K -stationarity enters locally through the stable-eigenvalue condition and globally as

a pathwise hypothesis verifiable by direct inspection of the simulated frozen-regime continuation; the primitive MPC condition (Appendix E, Lemma E.1) ensures no side-switching along the continuation but does not on its own rule out non-steady-state flat steps. Since $\{K_t\}$ is strictly monotone and converges to K_A^{cs} , Theorem 1 with $e(\Phi) = K(\Phi)$ yields $K(\Phi) = K(\Phi') \implies \Phi = \Phi'$ for all $\Phi, \Phi' \in \mathcal{M}(\Phi_0, A)$. ■

Proposition 7 makes the K -nullcline *sign-determining*: the graph $C = C_A^{Knull}(K)$ partitions the (K, C) plane into regions with $\Delta K > 0$ and $\Delta K < 0$ via $\Delta K = C_A^{Knull}(K) - C$. Assumption 2 rules out discrete-time jumps across this partition, so net investment cannot reverse sign along the conditional saddle; adding unique K -stationarity (the no-plateau condition) to rule out non-steady-state flat steps, capital dynamics are strictly monotone and each frozen-regime spell admits an exact one-dimensional representation indexed by K .

Proposition 8 makes $K : \mathcal{M}(\Phi_0, A) \rightarrow \mathbb{R}$ one-to-one, so an inverse map $\varphi_{\Phi_0, A}$ exists with $\Phi = \varphi_{\Phi_0, A}(K(\Phi))$ on $\mathcal{M}(\Phi_0, A)$.¹⁵ Any equilibrium object restricted to $\mathcal{M}(\Phi_0, A)$ then admits an exact single-valued representation as a function of (K, A) .

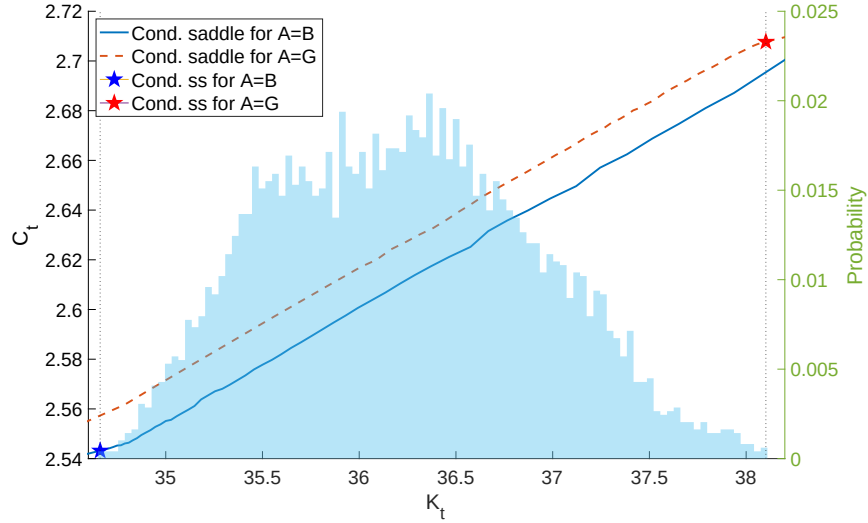


Figure 5: Conditional saddle paths in Krusell and Smith (1998)

Notes: The figure plots conditional saddle paths for $A = B$ (solid) and $A = G$ (dashed) implied by a canonical heterogeneous household business cycle model (Krusell and Smith, 1998). The histogram in the background plots the time-series distribution of the aggregate capital stock.

Figure 5 plots the computed conditional saddle paths in the (K, C) phase diagram.

¹⁵This is a path-specific notion of sufficiency: for a fixed (Φ_0, A) , K uniquely indexes states on the entire conditional saddle (globally *along the transition path*).

The model is solved globally using the repeated transition method, and the dynamics under each frozen aggregate state are simulated for 2,500 periods. Although the true endogenous state is Φ rather than K alone, the conditional saddle paths are strictly and monotonically ordered in K , providing clear computational support for the sufficiency result.

Resolving a narrow K -window shows the ergodic band that Figure 5 displays as a line to be “dusty”: within each spell the consecutive (K_t, C_t) points move monotonically in K , tracing an exact one-dimensional orbit, while ergodic mixing across spells produces the band observed at any given K .

From spell-wise to global sufficiency Proposition 8 is stated regime-by-regime: K uniquely indexes the conditional saddle $\mathcal{M}(\Phi_0, A)$ under a single frozen regime. The equilibrium path concatenates such one-dimensional segments, each exactly indexed by (K, A) within its spell, but the handoff between spells is governed by the full distributional state Φ_τ at the switch date, which (K_τ, A_τ) need not determine: different histories can arrive at the same (K_τ, A_τ) carrying different distributions. Spell-wise sufficiency is thus distinct from unconditional global sufficiency, which would require the same K to pin the same Φ regardless of path history. The following proposition bounds this across-spell discrepancy.

Observing the economy at a given K admits a “cloud” of distributions produced by different prior histories. When every frozen-regime map contracts, the current state is determined by the regime history up to an error decaying geometrically in its length, so two histories agreeing in recent periods produce nearby states.

Write $\Gamma_A := \Gamma_{\text{endo}}(\cdot, A)$. The maintained hypothesis, the substantive assumption of this subsection, is collected in the following definition.

Definition 7 (Uniform frozen-regime contraction).

The family $\{\Gamma_A\}_{A \in \{B, G\}}$ is a uniform frozen-regime contraction on $\mathcal{U}$ at rate $\lambda_c \in (0, 1)$ if $\mathcal{U} \subseteq \mathcal{X}$ is closed in the complete metric space $(\mathcal{X}, d)$, forward-invariant under every Γ_A (each Γ_A maps $\mathcal{U}$ into itself), of finite diameter $\text{diam}(\mathcal{U}) := \sup_{\Phi, \Phi' \in \mathcal{U}} d(\Phi, \Phi')$, and

$$d(\Gamma_A \Phi, \Gamma_A \Phi') \leq \lambda_c d(\Phi, \Phi') \quad \text{for all } \Phi, \Phi' \in \mathcal{U} \text{ and all } A.$$

Theorem 2 (Synchronization under frozen-regime contraction).

Let $\{\Gamma_A\}$ be a uniform frozen-regime contraction on $\mathcal{U}$ at rate λ_c (Definition 7). Then:

(i) (*Random fixed point: the history-dependent analogue of a contraction's unique fixed point*) For every regime history $\omega = (\dots, A_{t-2}, A_{t-1})$, the long-history limit (the state obtained by applying the history's maps from an ever more distant start)

$$\Phi^*(\omega) := \lim_{T \rightarrow \infty} (\Gamma_{A_{t-1}} \circ \Gamma_{A_{t-2}} \circ \dots \circ \Gamma_{A_{t-T}})(\Phi)$$

exists, is independent of $\Phi \in \mathcal{U}$, and is the unique state in $\mathcal{U}$ consistent with ω ; any trajectory initialized in $\mathcal{U}$ at date $t - T$ satisfies $d(\Phi_t, \Phi^*(\omega)) \leq \lambda_c^T \text{diam}(\mathcal{U})$.

(ii) (*Suffix synchronization*) If two histories ω, ω' agree over their most recent T periods, then

$$d(\Phi^*(\omega), \Phi^*(\omega')) \leq \lambda_c^T \text{diam}(\mathcal{U}).$$

Proof.

See Appendix C. ■

Part (i) is the random-fixed-point construction of [Schenk-Hoppé and Schmalfuß \(2001\)](#) specialized to the frozen-regime maps; part (ii) is its consequence for two states in the same- K cloud, and the cloud bound follows by averaging part (ii) over the common-suffix length. For the stationary symmetric two-state chain ($\pi_{BB} = \pi_{GG} = \rho_A$), let $q := \rho_A^2 + (1 - \rho_A)^2$ denote the probability that two independent histories agreeing at a date also agree one period earlier.¹⁶

Proposition 9 (Across-spell approximation bound).

Let $\{\Gamma_A\}$ be a uniform frozen-regime contraction on $\mathcal{U}$ at rate λ_c (Definition 7), and let $\Phi, \Phi' \in \mathcal{U}$ be the long-history limits of two regime histories that agree over their most recent $T \geq 1$ periods. Then

$$d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U}).$$

If, moreover, the two histories are drawn independently from the stationary symmetric chain, conditioned to share the current regime, then $T - 1$ is geometric with success probability $1 - q$ and

$$\mathbb{E}[d(\Phi, \Phi')] \leq \frac{(1 - q) \lambda_c}{1 - q \lambda_c} \text{diam}(\mathcal{U}).$$

¹⁶Stepping back one period, two independent histories still agree if both persist (ρ_A^2) or both switch ($(1 - \rho_A)^2$); for asymmetric chains, q is the backward agreement probability of the time-reversed chain.

Proof.

See Appendix C. ■

The diameter $\text{diam}(\mathcal{U})$ is finite under the borrowing constraint $a' \geq 0$ and compact idiosyncratic support. The right-hand side involves only the primitive set $\mathcal{U}$, not the same- K cloud itself, so the proposition is a genuine bound on the cloud rather than a contraction statement about it. The bound vanishes as $\rho_A \rightarrow 1$ (long spells make recent suffixes long) or $\lambda_c \rightarrow 0$ (fast within-spell convergence).

Uniform contraction is a substantive assumption The hypothesis that $\Gamma_{\text{endo}}(\cdot, A)$ contracts uniformly on $\mathcal{U}$ is not innocuous: the map embeds general-equilibrium price feedback, and establishing contraction-type properties for such operators is precisely the difficulty confronted by global existence and uniqueness analyses (Proehl, 2025). Locally, Proposition 10 anchors the contraction rate to the stable eigenvalue $\lambda_A^s \in (0, 1)$; globally the assumption is maintained, and the sample-based diagnostic below provides a lower-bound estimate, not a certification. The paper supplies no upper bound on λ_c : even $\lambda_c < 1$ is maintained rather than verified. No tension with the saddle structure arises: Γ_{endo} is the equilibrium law of motion after the forward-looking variable has been solved out, so the unstable direction of the (K, C) system is absent from the distributional dynamics.

Ergodic interpretation Theorem 2(i) licenses reading Proposition 9 on the ergodic set: every state observed along a long simulated path is, up to a geometrically small error, the long-history limit of its own realized history, the state that history alone drives the economy toward regardless of where it started. Every pair in the same- K cloud is therefore a pair of long-history limits, inheriting the pathwise bound $d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U})$ with T the realized common-suffix length. This applies at every date (deep within a spell or just after a switch) because the synchronization argument runs over histories rather than single spells.

The expectation formula, by contrast, is computed under the idealized model of two independent stationary histories, and two caveats attach: within-path pairs are dependent, and conditioning on a common K is a selection. The closed form is therefore a benchmark for the typical scale of the cloud rather than a bound on its realized diameter.

Theorem 2 leaves λ_c unspecified. The next proposition pins its value near the conditional steady state: the stable eigenvalue of the linearized frozen-regime dynamics, the same object governing the local saddle slope (Proposition 5), under the standard regularity of infinite-dimensional linearization (smoothness, a spectral gap, a stable manifold, and metric-norm agreement), stated with glosses in Appendix D.

Proposition 10 (Local contraction rate on the conditional saddle).

Near the conditional steady state, the frozen-regime dynamics on the conditional saddle contract distances at the geometric rate $|\lambda_A^s|$, where $\lambda_A^s \in (0, 1)$ is the stable eigenvalue of the linearized frozen-regime dynamics; in aggregate-capital units,

$$K_{t+1} - K_A^{cs} = \lambda_A^s (K_t - K_A^{cs}) + o(|K_t - K_A^{cs}|).$$

Proof.

See Appendix D. ■

The full statement on the distributional state space, its regularity conditions, and the proof are in Appendix D. Directions orthogonal to the saddle may contract at different rates, so the global rate λ_c in Theorem 2 could exceed $|\lambda_A^s|$; the proposition delivers the primitive anchor, the same eigenvalue that drives the slope-ratio diagnostic (Proposition 5) and disciplines the local contraction in Theorem 2 and Proposition 9. A certified uniform upper bound on λ_c is left to future work.

A sample-based diagnostic for λ_c The global uniform λ_c in Theorem 2 bounds contraction across a cloud of distributional states at the same K reached from different histories, which need not lie on a single conditional saddle. The diagnostic below can show the map contracts no faster than some observed rate; it cannot certify that it contracts at least that fast for every pair of states, which is what the bound needs. Since the across-spell bound uses λ_c as an *upper* bound on the per-step rate, a certified verification of Proposition 9 requires a certified upper bound on λ_c , via either a Lipschitz constant for Γ_{endo} on $\mathcal{U}$ (model-specific) or a statistical confidence bound on sample estimates (standard but tedious in function spaces); this is beyond the scope of the paper and left to future work. Instead I propose a sample-based *diagnostic* that provides a lower bound on λ_c and a reality check against the local anchor $|\lambda_A^s|$.

The diagnostic draws N pairs of distributional states at common K from the simulated equilibrium, computes their one-step contraction ratios r_i , and reports

$\hat{\lambda}_c^N := \max_i r_i$. Since $\hat{\lambda}_c^N \leq \lambda_c$, this is a *lower* bound, and substituting it in the across-spell formula gives an optimistic approximation; the along-saddle anchor $|\lambda_A^s|$ (Proposition 10) serves as a reality check, since close agreement is indirect evidence that the true λ_c is not much larger.

Under the Krusell and Smith (1998) calibration, quarterly regime persistence is $\rho_A = \pi_{AA} = 0.875$ (average 8-quarter spells) and the diagnostic yields $\hat{\lambda}_c^N \approx 0.964$, with the local anchor $|\lambda_A^s| \approx 0.963$ agreeing to two decimals. With $q = \rho_A^2 + (1 - \rho_A)^2 \approx 0.781$, substituting $\hat{\lambda}_c^N$ into the benchmark formula of Proposition 9 gives $\mathbb{E}[\lambda_c^T] \approx 0.85$ as the typical contraction factor multiplying $\text{diam}(\mathcal{U})$ under the independent-histories benchmark, an optimistic estimate on both counts. The direct quantitative check, comparing the globally solved K' at each spell boundary against the prediction from tracking only (K, A) , gives an average discrepancy of 2.5×10^{-4} in log terms (1.3% of one standard deviation of $\ln K$). This measures a mean forecast error in K' rather than a worst-case distance between distributions, so it is not directly comparable to the uniform distributional bound above; it separately documents that the across-spell error is small in this calibration.

State reduction versus functional-form approximation The Krusell and Smith (1998) approach is often labeled near-rational (Akerlof and Yellen, 1985), but this conflates two sources of approximation the conditional-saddle structure separates: within an orbit conditioning on K is exact and only the functional form of the law of motion approximates, while across histories a residual state-reduction error remains—small in this calibration (Proposition 9)—which the framework identifies rather than assumes away.

Connection to block recursive equilibrium The conditional sufficiency of K relates to *block recursive equilibrium* (BRE), where aggregate prices and quantities follow from a low-dimensional state (Menzio and Shi, 2010, 2011; Schaal, 2017). In directed-search environments block recursivity holds by construction; here it emerges spell-wise from the saddle geometry, with prices, quantities, and Φ recoverable from K along a given conditional saddle. This is weaker than conventional (K, A) block recursivity: recovery also requires the orbit identity fixed by the generating history, since different histories reach the same (K, A) with different distributions, agreeing only approximately under a common recent history (Proposition 9).

Parametric form of the conditional saddle paths The indexing function K does not pin down the form of the law of motion. Theorem 1 and Proposition 8 nonetheless justify the repeated transition method (Lee, 2026a), which uses K as the conditional sufficient statistic in heterogeneous-agent models.

Verifying monotonicity in practice The framework also yields a practical deliverable: a cheap, computable check of the monotonicity hypothesis of Theorem 1 (Assumption 2 with unique K -stationarity)—one any solution method can run to certify scalar-state sufficiency, or flag its failure, before relying on it. It is verifiable two ways, both in the supplement. A *perfect-foresight diagnostic* (Appendix G) transports K -monotonicity from the deterministic transition path, which practitioners already compute, to rational expectations along a regular belief homotopy, with nullcline invariance the immovable boundary the path cannot cross without violating no-plateau. Independently, primitive bounds on household MPCs and price feedback control side-switching under rational expectations, yielding exact within-spell K -sufficiency under log utility and Cobb–Douglas production; the global check is verified along a computed continuation rather than from primitives alone, with two conditions maintained explicitly—a pathwise finite-difference bound on aggregate consumption and the residual no-plateau condition (Appendix E). The logic extends to bonds, GHH labor, and multiple predetermined states, its boundary the lumpy-investment economy where nullcline invariance fails and monotonicity becomes a testable diagnostic that tracks where scalar-state accuracy degrades (Appendix J).

4.2 Search and matching: comparative statics

The conditional saddle framework applies to any recursive competitive equilibrium with a predetermined endogenous state, not only neoclassical growth models. I apply it to a Diamond–Mortensen–Pissarides search-and-matching economy, where unemployment is the predetermined state and the downward-sloping Beveridge-curve loop in (u, v) data emerges from transitions across conditional saddle paths. The slope asymmetry of Lemma 1 and Proposition 5 acquires a labor-market reading: the surplus compression that the fundamental-surplus analysis of Ljungqvist and Sargent (2017) places at the heart of the Shimer puzzle manifests as differential saddle steepness. Comparative statics on unemployment benefits then reveal a non-monotone volatility

relationship, traceable to convergence of the regime-specific conditional steady states; extensions are in Appendix I.

The DMP environment Consider a discrete-time stochastic search-and-matching economy.¹⁷ Aggregate productivity follows a Markov chain with extreme states $z \in \{G, B\}$. The aggregate state is $S = [u_{-1}, z]$, with unemployment $u_{-1} = 1 - n_{-1}$ the sole predetermined endogenous state. Employment evolves according to

$$n = (1 - \lambda) n_{-1} + q(\theta) v, \quad (32)$$

where $\lambda \in (0, 1)$ is the separation rate, $\theta = v/u$ is labor market tightness ($u = 1 - n$ is the current unemployment pool, the predetermined state being inherited $u_{-1} = 1 - n_{-1}$), and $q(\theta) = m_0 \theta^{-\xi}$ is the Cobb–Douglas vacancy-filling probability. Tightness $\theta(S)$, and hence vacancies $v(S) = \theta(S) u(S)$, is the jump variable, pinned down by free entry:

$$\frac{\kappa}{q(S)} = \beta \mathbb{E} \left[\left(\frac{c(S)}{c(S')} \right)^\sigma \left(z' - w(S') + (1 - \lambda) \frac{\kappa}{q(S')} \right) \middle| S \right], \quad (33)$$

where κ is the posting cost, $w(S)$ the Nash-bargained wage, and $c(S) = n(S) z - \kappa v(S) + (1 - n(S)) b$ aggregate consumption. Under the hybrid timing convention, current-period matches produce immediately while free entry values a vacancy from the next period onward (Appendix I).

The conditional saddle path $\mathcal{M}(u_0, z)$ is the orbit $\{(u_t, v_t)\}_{t \geq 0}$ generated by (32) and the equilibrium vacancy function $v(\cdot, z)$ under frozen productivity z . Its limit point is $(u_z^{\text{cs}}, v_z^{\text{cs}})$.

Slope asymmetry and the Beveridge curve Both conditional saddle paths in (u_{-1}, v) space slope upward, so state dependence is sign-preserving as in the RBC benchmark; but the slopes differ markedly, steep under G where matches are profitable and nearly flat under B where the flow surplus is low (Figure 6a).

The mechanism is free entry (33): each unemployed worker is a potential match whose value depends on productivity. Under high z the flow surplus $z - w$ is large, so firms respond aggressively to a larger unemployment pool, producing a steep locus. Under low z the surplus is compressed, since with the outside option b fixed the Nash

¹⁷The full model specification and calibration are in Appendix I. I adopt a standard formulation following Shimer (2005) with Nash-bargained wages, exogenous separation, and Cobb–Douglas matching. The computation uses an 11-state Tauchen discretization of log TFP; for expositional simplicity, I present the two extreme states as G and B .

wage falls less than one-for-one with productivity, squeezing $z - w$ (Hagedorn and Manovskii, 2008; Ljungqvist and Sargent, 2017), and posting barely responds. The slope asymmetry is the visual signature of the surplus compression underlying the unemployment-volatility puzzle of Shimer (2005).

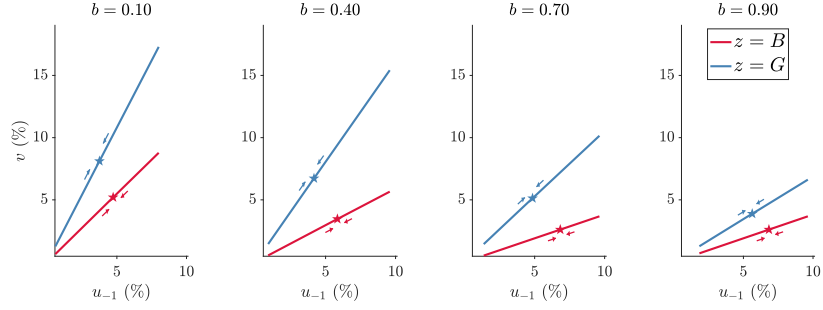
This asymmetry generates the Beveridge-curve “loop” in empirical (u, v) data. From near the G steady state, a negative shock switches the economy onto the flat B saddle: vacancies collapse on impact and barely adjust as unemployment accumulates. Upon recovery it jumps back onto the steep G saddle at high inherited u_{-1} , driving a sharp vacancy spike. The counterclockwise rotation through recessions and recoveries is thus across-saddle jump dynamics.

Comparative statics of the benefit level Ergodic unemployment volatility is hump-shaped in the benefit level b . The operative diagnostic is the gap between regime-specific conditional steady states, $u_B^{\text{cs}}(b) - u_G^{\text{cs}}(b)$, not slope asymmetry alone: at high b , u_B^{cs} saturates while u_G^{cs} keeps rising, so the gap collapses, and the synchronization contraction of Theorem 2(i) anchors each regime’s ergodic states near its conditional steady state, compressing the range over which unemployment can fluctuate.¹⁸ This is the same fundamental-surplus channel through which Ljungqvist and Sargent (2017) show the benefit level governs amplification across DMP variants; the saddle geometry adds the steady-state-gap diagnostic that makes the non-monotonicity visible. Robustness is in Appendix I.

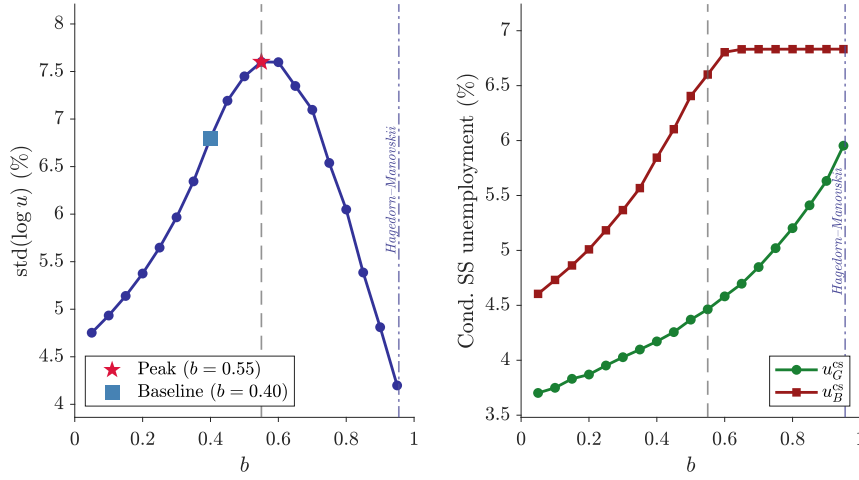
The Nash wage $w = (1 - \eta)b + \eta(z + \kappa\theta)$ rises in b , so raising b compresses the surplus $z - w$, proportionally more in the B regime where $z - b$ is already small. Figure 6a reports the conditional saddle paths for $b \in \{0.10, 0.40, 0.70, 0.90\}$: the G saddle is everywhere steeper than the B saddle, and as b rises the deepening compression flattens both loci while pushing the conditional steady states toward higher unemployment. But volatility is non-monotone (Figure 6b): it peaks at $b \approx 0.55$ ($\text{std}(\log u) \approx 7.6\%$, with a nearly flat top through $b \approx 0.60$) and then *declines*. The peak location is calibration-specific, and this is a ceteris-paribus comparative static: the amplification in Hagedorn and Manovskii (2008) at $b \approx 0.955$ works through their joint recalibration of the bargaining weight and posting cost, not the benefit level alone, so the hump neither contradicts nor overturns their calibration.¹⁹

¹⁸The theorem’s uniform-contraction hypothesis is maintained for the scalar-state DMP frozen-regime map, where it is directly checkable.

¹⁹In this parameterization, simulated volatility is about 4.2% at $b = 0.955$ versus 6.8% at the



(a) Conditional saddle paths



(b) Unemployment volatility (left) and conditional steady-state unemployment (right)

Figure 6: Comparative statics of b

Notes: Panel (a): conditional saddle paths in (u_{-1}, v) space for $b \in \{0.10, 0.40, 0.70, 0.90\}$; stars mark conditional steady states. Panel (b), left: $\text{std}(\log u)$ over 19 values of b ; the star marks the peak ($b \approx 0.55$), the square marks the baseline ($b = 0.40$), and the dash-dotted vertical line marks the [Hagedorn and Manovskii \(2008\)](#) replacement rate ($b = 0.955$). Panel (b), right: conditional steady-state unemployment under frozen good (u_G^{cs} , green) and bad (u_B^{cs} , red) TFP, extracted from the stochastic policy functions. Both the gap $u_B^{\text{cs}} - u_G^{\text{cs}}$ and simulated volatility are hump-shaped in b .

The right subplot of Figure 6b traces the mechanism: u_G^{cs} climbs steadily while u_B^{cs} rises faster and then *saturates* near 6.8% (agents expect TFP recovery, so vacancies persist under bad productivity). The gap widens from under 1 p.p. at $b = 0.05$ to about 2.2 p.p. near $b \approx 0.60$, then collapses, so gap and volatility are hump-shaped in b , peaking together.

baseline $b = 0.40$; the saddle diagnostic identifies steady-state convergence as the mechanism.

Sensitivity to wage setting and matching elasticity The hump is robust across wage-setting rules and for matching elasticities up to the baseline (Appendix I, Figure I.1). Under Nash bargaining with $\xi \in \{0.40, 0.50, 0.635, 0.80\}$ (Hosios $\eta = \xi$ maintained), the peak shifts leftward as ξ rises and volatility is nearly flat at $\xi = 0.80$; under present-wage bargaining, which removes the tightness-to-wage feedback, the hump persists with peak near $b \approx 0.93$. The pattern reflects surplus geometry, not a specific wage-feedback channel.

5 Concluding remarks

Conditional saddle paths, equilibrium trajectories under frozen exogenous regimes, bring the geometric clarity of deterministic phase diagrams to stochastic environments. Their slopes encode state dependence: differential tilt across saddles is the geometric signature of asymmetric shock responses. Their monotonicity delivers dimension reduction, since a strictly monotone aggregate variable exactly indexes the equilibrium state within each frozen-regime spell. Persistent spells, exact within-spell sufficiency, and a synchronization property under a maintained contraction condition (the economy forgets its regime history at a geometric rate) identify the mechanism by which the conditional result would translate into near-global accuracy. The bound this mechanism delivers is loose in the calibration examined, so we document the near-exact accuracy directly rather than through the bound; together they give a structural account of why scalar-state methods work.

The canonical RBC model establishes the basic geometry and reveals that state-dependent response shapes arise from distance along the saddle even when slopes are nearly equal. Two applications then illustrate the framework's reach. The heterogeneous-household model of [Krusell and Smith \(1998\)](#) demonstrates that strict monotonicity along conditional saddles delivers exact within-spell dimension reduction: locally under standard regularity, and globally on an explicit region whose conditions can be checked from calibrated parameters, with one residual assumption stated rather than hidden; the across-spell error is bounded, under the maintained uniform-contraction condition, by how quickly the economy forgets its regime history. A non-concave production illustration shows the no-crossing condition is a genuine geometric restriction, and a lumpy-investment experiment confirms the diagnostic tracks where scalar-state accuracy degrades. The search-and-matching application

shows the framework’s reach beyond neoclassical growth: the Beveridge-curve loop emerges from transitions across regime-specific saddle loci, surplus compression under low productivity manifests as differential saddle steepness, and the geometry provides a direct diagnostic for the non-monotone relationship between benefit generosity and unemployment volatility.

More broadly, conditional saddles offer a complementary lens on stochastic models: a language for interpreting nonlinear dynamics, for assessing when scalar-state reductions are exact on the equilibrium path rather than merely accurate, and for characterizing across-path variation as a structural feature of the geometry. A promising direction is to use these objects to sharpen empirical restrictions on state dependence and to discipline the design of state-contingent stabilization policies.

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